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A last-mile delivery system for underground logistics with “self-pickup +” and “home-entry +” modes

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ABSTRACT

Underground logistics systems (ULSs) are emerging low-carbon, energy-saving, and ground-space-saving logistics and transportation approaches that rely on underground tunnels/pipelines for freight transportation. Thus, they will effectively solve the urban logistics dilemma caused by the rapid growth of express parcel volumes in China, especially concerning last-mile delivery in cities, which is currently one of the most expensive and least efficient parts of the entire logistics and transportation process. We designed a three-level (i.e., city, regional, and neighborhood) last-mile delivery system (LDS) for express parcel transportation according to a metro-based ULS (M – ULS). We categorized the LDS into two modes: “self-pickup +” and “home-entry +.” According to different needs, we then applied the system to five different scenario–mode combinations: “self-pickup + metro stations,” “self-pickup + shopping malls,” “self-pickup + communities,” “home-entry + communities,” and “home-entry + office buildings.” Finally, using an urban sub-center in China as a case study, we analyzed efficiency and emission reduction of LDS in the five scenario–mode combinations with different freight volumes and verified the validity of the proposed system. This system has great potential for solving last-mile delivery challenges and provides new ideas and technical solutions for planning and designing future ULSs.

1. Introduction

Urban logistics is often seen as an equivalent to parcel deliveries or courier-, express-, and parcel providers (Nuzzolo et al., 2016; Shao et al., 2019; Yang et al., 2016; Zhan et al., 2023), and competition between these for available capacity on urban road networks has triggered a series of new challenges such as traffic congestion and urban environmental pollution (Chen et al., 2018a; Kummer et al., 2020; Marcucci et al., 2017). In recent decades, with the rapid growth of e-commerce and online shopping, Chinese express delivery volumes have markedly increased from 110 million pieces in 2010 to over 110.58 billion pieces in 2022 (State Post Bureau of The People's Republic of China, 2022). Consequently, China has become the country with the highest number of express delivery parcels in the world (Ding et al., 2020; Jiang et al., 2023). Among them, the delivery volumes of intra-city express service in 2022 was 12.08 billion (State Post Bureau of The People's Republic of

China, 2022). However, many cities in China lack sufficient developments in infrastructure and transportation planning to sustain the growing demand for urban express parcels transportation. The urban underground space, as a component of an urban three-dimensional space (including spaces above ground, on the ground, and below ground), offers a broader transportation space that can alleviate surface traffic congestion, particularly in the transportation of surface logistics, as it does not occupy surface roads (Bobilev, 2009; Bobilev & Sterling, 2016; Broere, 2016; Cui et al., 2021; Guo et al., 2021; Lin et al., 2022). Underground logistics systems (ULSs) are an emerging logistics and transportation approach that rely on underground tunnels or pipelines for freight transportation and offer low-carbon, energy-saving, and ground space-saving benefits (Chen et al., 2017; Chen et al., 2020; Guo et al., 2021; Zheng et al., 2022). Thus, they will be an effective solution for the urban logistics dilemma presented by the increasing number of express parcels owing to the increasing e-commerce development in

Abbreviations: ULS, Underground logistics system; LDS, Last-mile delivery system; AGV, Automatic guided vehicle; ULDZ, Urban logistics distribution zone; MFSL, Metro freight station location; ULTTC, Urban logistics terminal transit center; LDT, Last-mile delivery terminal; LDTF, Last-mile delivery tunnel facility; SL, Subway Lines; SULTF, Specialized Underground Logistics Tunnel Facilities.

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China.

Transferring express parcel transportation to underground spaces and utilizing the depth and breadth of urban underground space to establish an ULS can overcome the bottleneck of existing urban logistics systems and improve logistics and transportation efficiency (Chen et al., 2017; Zhao et al., 2018; Zheng et al., 2020). A ULS is a transportation and supply system for fully automated freight transportation using a carrier or medium such as an automatic guided vehicle (AGV) or capsule trolley—which is powered by sustainable energy—to ultimately distribute freights to various terminals individually or in bulk in a confined space such as an underground tunnel or a pipeline (Di et al., 2022; Vernimmen et al., 2007; Zhao et al., 2021). Therefore, a ULS does not occupy surface roads, is physically isolated from external conditions, and can use fully automated technology to guarantee reliable freight transportation, while effectively reducing urban pollution (Hai et al., 2020; He et al., 2020; Wei et al., 2020).

ULSs are independent of network transportation and have the advantages of utilizing a separate management control system and maximizing freight capacity; simultaneously, as an independent underground logistics distribution network, they do not interfere with other transportation systems and only occupy underground closed spaces for automated equipment operations and freight efficiency (Chen et al., 2018b; Chen et al., 2020; Guo et al., 2021; Hai et al., 2020). Freight transport and independent ULS distribution networks in cities are generally divided into three main distribution nodes: transportation of freights from (1) logistics parks outside the city (logistics center) to (2) urban distribution centers (warehouses), and then to (3) the terminal outlets or customers (represented in Fig. 1) (Guo et al., 2021; He et al., 2020; Kim et al., 2022; Zhang et al., 2022). ULSs consist of mainline and feeder underground lines. The mainline underground lines use large-diameter tunnel cross-sections that allow for the transportation of containers, independent rights-of-way, and fixed hours of operation. The feeder underground lines use smaller tunnel cross-sections with less freight capacity and AGVs and other delivery vehicles to achieve turnkey services (Fig. 1). Notably, the underground freight transport technologies used for mainline and feeder underground lines can meet transportation requirements. ULSs also include nodes such as terminals and logistics centers with different freight-handling capacities to serve different regions and customers.

As the most expensive and least efficient part of urban logistics and transportation process (Gevaers et al., 2014; Ouyang et al., 2023), last-mile deliveries have become the focus of research on ULS planning and design. Last-mile delivery is defined as the design and provision of delivery services from a public delivery node, such as a depot, to customers' specified destinations (Ouyang et al., 2023; Ranieri et al., 2018;

Wang et al., 2016; Wang, 2019). The “last-mile” refers to the time a courier package leaves the transportation system; that is, the last step in the delivery process. It is the final leg of business-to-consumer package delivery to consumers' homes, cluster/collection points (Aized and Srai, 2014; Chen et al., 2021a; Perboli et al., 2018; Yu et al., 2021), or storage until the recipient picks it up or forwards it to another address. It links online ordering and product delivery (Esper et al., 2003; Guo et al., 2019; Patella et al., 2020; Zhou et al., 2020). Last-mile delivery is considered one of the most expensive parts of the supply chain, accounting for 13–75 % of the total supply chain cost, and involves a set of activities that are necessary for the delivery process from the last point of transit to the final drop-off point in the delivery chain. Therefore, this study aimed to solve problems plaguing the last-mile delivery of urban express parcels by designing a last-mile delivery system (LDS) based on a ULS.

Most of the current research on ULS, metro-based underground logistics system (M-ULS), and underground container logistics has focused on planning and application in new cities and districts (Guo et al., 2021), modeling of network planning (Hu et al., 2022a; Pan et al., 2019), mechanisms for network expansion (Chen et al., 2023), effects on urban freight traffic (Hai et al., 2020), benefits and long-term development (Hu et al., 2022b; Pan et al., 2023), joint dispatch of shipment equipment (Liang et al., 2022), and other macroscopic aspects. Only part of the literature focuses on microscopic research. For example, Fan et al. (2020) describe the internal layout and floor space of the port-convergence-station for the ULS to solve the container transportation problem. Hu et al. (2021) designed an indoor layout scheme for a specific metro interchange station with logistics functions. Rezaeifar et al. (2022) developed a framework for optimizing the capacity of intermodal underground logistics transportation terminals through the discrete event simulation model. Mo et al. (2023) introduced the vehicle routing problem in underground logistics systems to model how to find the optimal vehicle routing and freight transfer solution. However, these studies did not focus on the design of ULS for last-mile delivery in urban logistics. Last-mile delivery can be divided into two modes: “self-pickup +” and “home-entry +.” To meet the needs of different scenarios, we aimed to design an LDS for last-mile delivery using ULS in two modes, and validate the feasibility of the workflow for each of them in an urban sub-center of a Chinese city. The novel contributions of this study to existing literature are as follows:

- It presents an LDS based on a metro-based urban underground logistics system (M – ULS), designed based on existing research on ULSs. This system can achieve transportation of express parcels at city, regional, and neighborhood levels, focusing especially on the

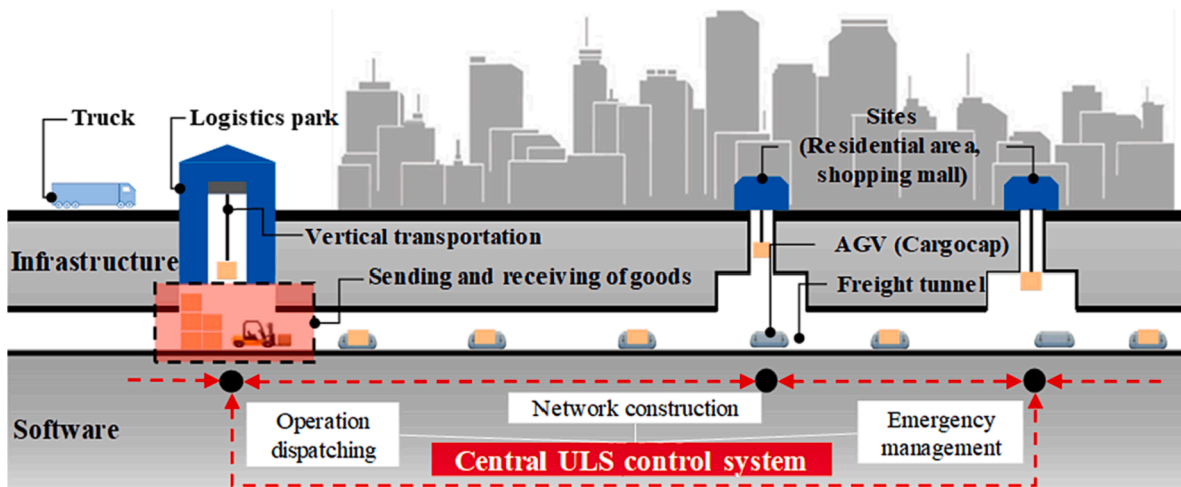


Fig. 1. Schematic of a typical mode of operation for ULSs (Guo et al., 2021).

structure of the distribution system at the neighborhood level (the last-mile delivery level). It also presents a distribution process.

- This system comprises two major modes, “self-pickup+” and “home-entry +,” for different last-mile delivery scenarios using a ULS. The “self-pickup+” mode was designed for three scenarios, metro stations, communities, and shopping malls, while the “home-entry +” mode was designed for two scenarios, office buildings and communities.
- To assess the feasibility of the ULS in a case study, we selected one metro freight station and five LDSs in a city’s sub-center in China and calculated parameters for forward logistics and reverse logistics to determine distribution time.

This paper is organized as follows: Section 2 details the design of the ULS for express parcel delivery. Section 3 describes the construction of last-mile delivery modes for the different “self-pickup +” and “home-entry +” mode scenarios. Section 4 presents the case study and its results. Section 5 compares the LDS constructed in this study with current last-mile delivery, and outlines the advantages of the former. Section 6 summarizes the findings and limitations of the study.

2. Structural design of an underground logistics system for express parcel transportation

Transportation of freights via urban rail transit systems has gradually gained research attention and is considered an essential means of constructing urban ULSs (Chen et al., 2023; Guo et al., 2021; Villa and Monzón, 2021). The availability of the metro system during off-peak hours can be leveraged for urban freight transport to avoid loss of transportation capacity (Ma et al., 2022; Zhao et al., 2021; Zuo et al., 2023). Here, we designed a ULS that utilizes the metro to transport express parcels in the city, based on the existing ULS, referred to here as the M – ULS, based on existing research (Guo et al., 2021; Zhao et al., 2021; Zuo et al., 2023). This ULS for the transportation of express parcels consists of several components (Fig. 2), the functions of which are described in detail below.

- Urban Logistics Distribution Zone (ULDZ): Most of the express parcels in the urban underground logistics system come from ULDZs, which are generalized to areas with intensive logistics activities such as ports, airports, and highway terminals. Most ULDZs are located in the outskirts of the city and cannot be easily connected to the metro network in the urban areas. Therefore, the ULDZs were renovated to allow them to serve as underground logistics terminals; additionally, the ULDZs were connected to the metro network within the urban area through the construction of additional rail-based lines.
- Metro Freight Station Location (MFSL): The MFSL is in the original metro station and provides an increased freight platform for loading and unloading metro locomotives delivering express parcels, in combination with a series of horizontal/vertical facilities for handling, centralized parcel processing (sorting, warehousing, consolidation, loading and consolidation of the parcels), and transshipment (to ULDZs or ULTTCs).
- Underground Logistics Terminal Transit Center (ULTTC): ULTTCs enable the transfer of express parcels between MFSLs and last-mile delivery terminals (LDTs). Further, express parcels are loaded, unloaded, sorted, and transferred here, but not stored.
- LDTs: LDTs are where the express parcel delivery step is performed. The delivery modes are divided into “self-pickup +” and “home-entry +” and, according to different scenario requirements, can be applied to five different scenarios: “self-pickup + subway stations,” “self-pickup + shopping malls,” “self-pickup + communities,” “home-entry + communities,” and “home-entry + office building.”.
- Last-mile Delivery Tunnel Facility (LDTF): This distributes express parcels from ULTTCs to LDTs via pallet transportation tools and corresponding two-way last-mile delivery tunnels.
- Subway Lines (SL) and Specialized Underground Logistics Tunnel Facilities (SULTF): SL achieves the transport of express parcels from the ULDZ to the MFSL and the MFSL. Transportation involves the transformation of passenger subway cars to enable the transport of larger quantities of express parcels. SULTF achieves the transfer of express parcels between the MFSL and ULTTC via small locomotive transport.

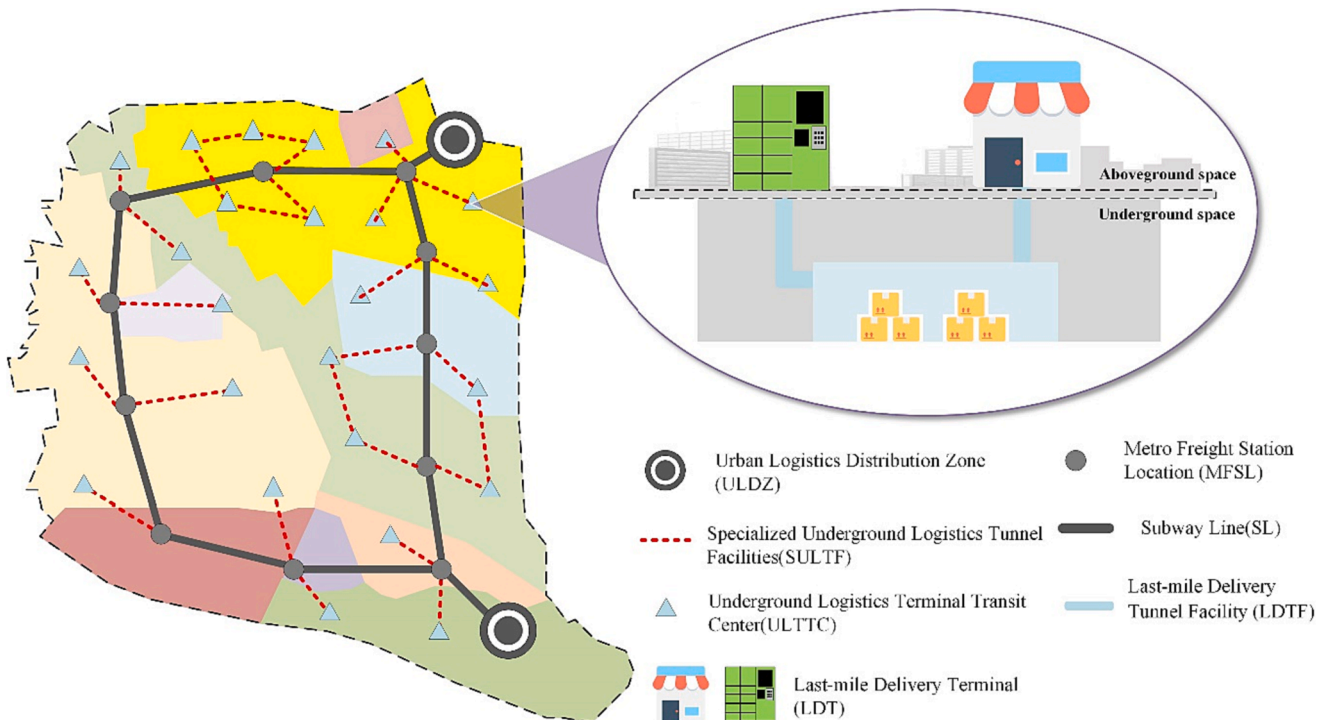


Fig. 2. Schematic diagram of the underground logistics system for express parcel delivery.

The proposed system for the transportation of express parcels can be performed at the city (from the ULDZs to the MFSs within the city), regional (via metro freight station transport between the MFSs and the ULTCs), and neighborhood levels (from MFSs to LDTs). The three-level express parcel delivery process (Fig. 3) is detailed below.

- **City level:** First, express parcels arrive at the ULDZ through various modes of transportation and are sorted, then they are transferred to the underground transportation hub located in the ULDZ and are transported to the MFSs in the urban area through the SL network.
- **Regional level:** The SL transports express parcels in the MFSL to the ULTC, while the SULTF transports express parcels from the MFSL to the ULTC.
- **Neighborhood level:** Distribution is divided into “self-pickup +” and “home-entry +”, according to the scenario. Express parcels located in the MFSs and ULTCs are transported to the LDTs. The express parcels are then collected by customers or directly delivered to the home of the customer.

The above transportation process can be summarized as “city level → regional level → neighborhood level,” representing the forward transportation of express parcels, known as “forward logistics.” The definition of “reverse logistics” in this study refers specifically to the reverse transportation of express parcels. The transportation process described as “neighborhood level → regional level → city level” is reverse transportation, defined as “reverse logistics.”

3. Design of a last-mile delivery system for underground logistics system

According to the “self-pickup +” and “home-entry +” modes, two categories were constructed for five different scenarios: “self-pickup + metro stations,” “self-pickup + shopping malls,” “self-pickup + communities,” “home-entry + communities,” and “home-entry + office

buildings.” Their definitions and characteristics are further discussed below.

- **“Self-pickup+” model:** This mode concerns the transportation of express parcels from the MFSL or ULTC through the LDTF to the LDT, wherein the last step is the customer arriving at the LDT to complete the delivery. In last-mile delivery, the “self-pickup +” mode involves intelligent express parcel cabinets, collection points, property collection, and convenience store collection. The design principle comprises few express parcels concentrated at an LDT, which are then collected by the customer, thereby solving express parcels delivery challenges. This mode is suitable for when there are many customers within the LDTs’ service range. An advantage of the “self-pickup+” mode is that it shortens the last-mile delivery path, reduces the number of last-mile deliveries, and dramatically improves distribution efficiency.
- **“Home-entry+” model:** Compared with the “self-pickup+” model, the “home-entry+” model involves the transport of express parcels through the LDTF directly to each floor of a building or each customer’s household. The “home-entry+” mode LDTs take the form of intelligent couriers. The design principle is to transport express parcels directly to each customer to complete the delivery. The “home-entry +” mode is suitable for customers with requirements for high express parcel delivery efficiency and who require privacy. An advantage of “home-entry+” mode is that parcels are delivered accurately and quickly, therefore customers can receive express parcels without leaving their homes, providing high quality service and improving last-mile delivery efficiency.

3.1. Design of the “self-pickup+” mode

The “self-pickup+” mode was designed for three scenarios: metro stations, communities, and shopping malls, which are discussed in detail

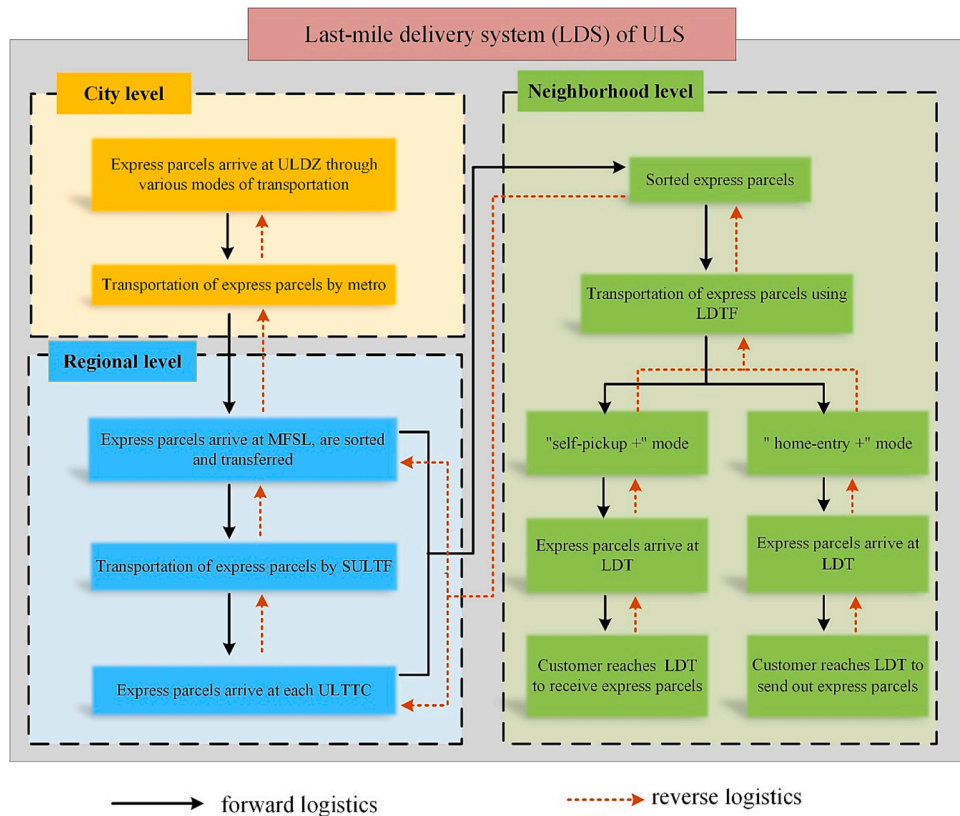


Fig. 3. Transportation flow chart of the underground logistics system for express parcel delivery at three levels.

as follows.

3.1.1. “Self-pickup + metro stations” mode

The “self-pickup + metro stations” model involves the transport of express parcels from the MFSL to the SULTF, then to the end of the network (the LDT) located above ground at the entrance to the metro station. Then, the customer collects the parcel from the LDT to complete the delivery process, as shown in Fig. 4. The “self-pickup + metro station” model works as follows: (1) Unloading and temporary storage of express parcels at MFSLs; (2) Sorting of unloaded express parcels according to the customer information carried on the waybill and those belonging to the LDT at the metro station are removed; (3) Delivery of express parcels to the LDTs at the exits of metro stations by the LDTFs; (4) Collection of express parcels or the sending or returning of parcels by customers at their own pace when passing through a station or leaving a station at an LDT.

3.1.2. “Self-pickup + communities” mode

The “self-pickup + communities” mode involves the transportation of express parcels from the MFSL located in the community to

underground through the LDTF located at the LDT, followed by the collection of the express parcels by customers from the communities to complete the delivery at the designated LDTs, as shown in Fig. 5. The “self-pickup + communities” model works as follows: (1) Express parcels are sorted according to different residential LDTs using waybill customer information and are transported to the ULTTC; (2) Express parcels are then delivered to the LDTs in the corresponding community via LDTFs; (3) Customers in the community go to the designated LDT to collect and send express parcels through this LDT.

3.1.3. “Self-pickup + shopping malls” mode

The “self-pickup + shopping malls” mode involves the transport of express parcels from MFSLs located below malls, which are then transported through LDTFs to LDTs located in shopping malls. Customers then collect the parcels from the shopping mall at the designated LDTs to complete the delivery, as shown in Fig. 6. The “self-pickup + shopping malls” model works as follows: (1) Express parcels are sorted according to the shopping malls of different LDTs using waybill customer information from the MFSL; (2) The express parcels are delivered to LDTs at the corresponding shopping malls via LDTFs; (3) Customers collect their

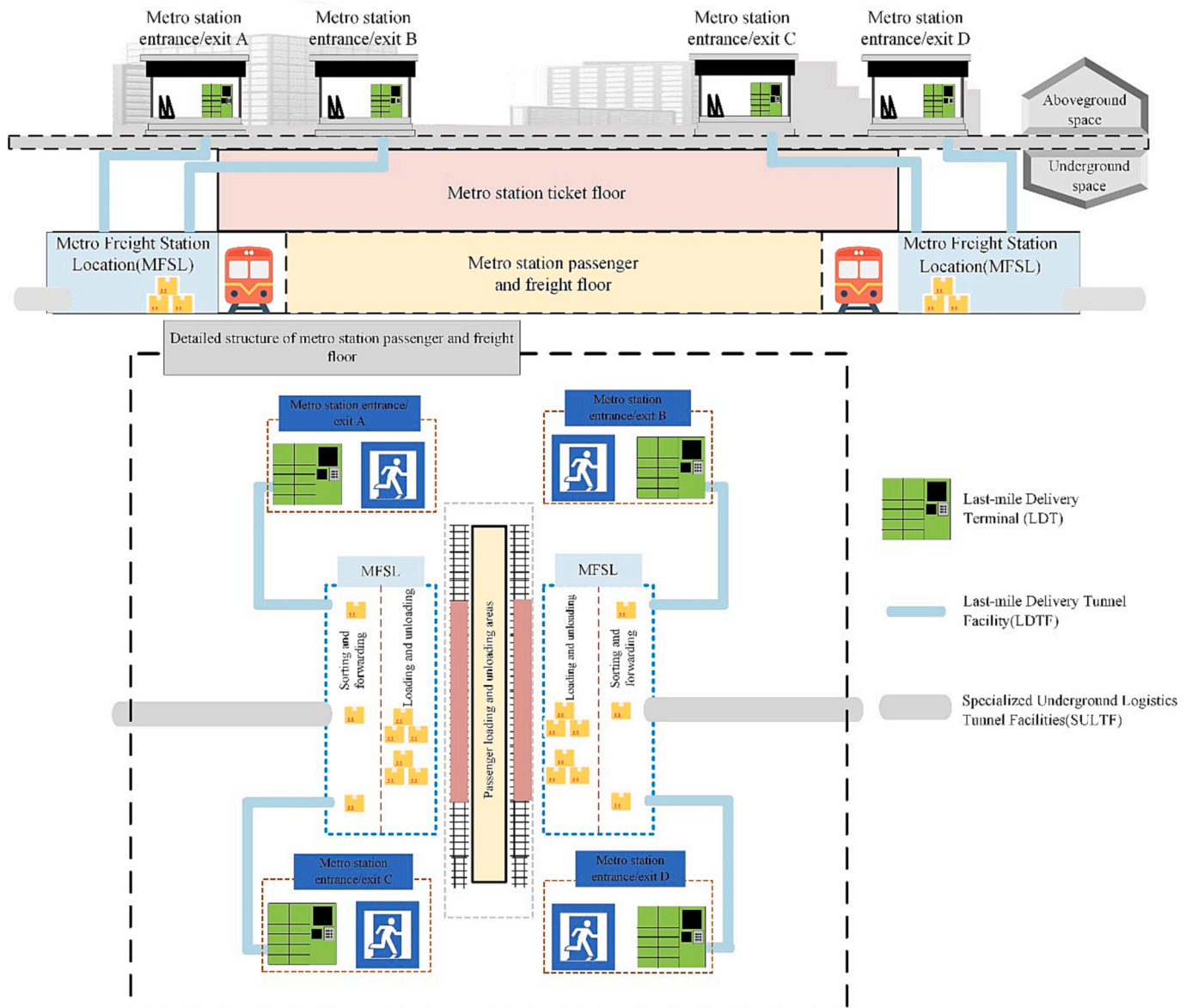


Fig. 4. Scheme design of the “self-pickup + metro stations” mode.

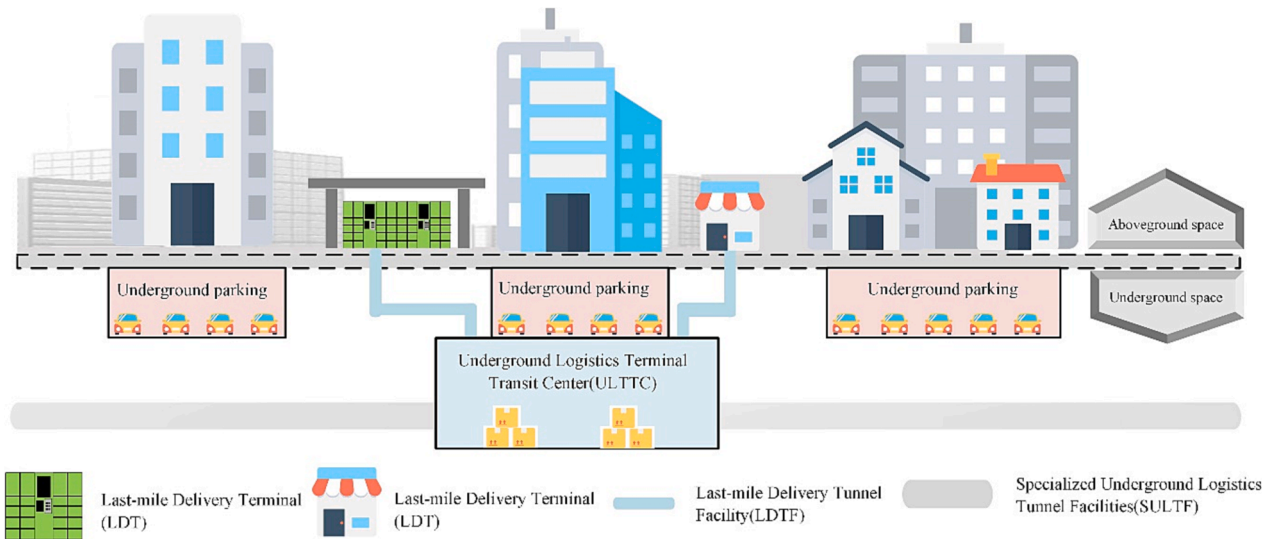


Fig. 5. Schematic design of the “self-pickup + communities” mode.

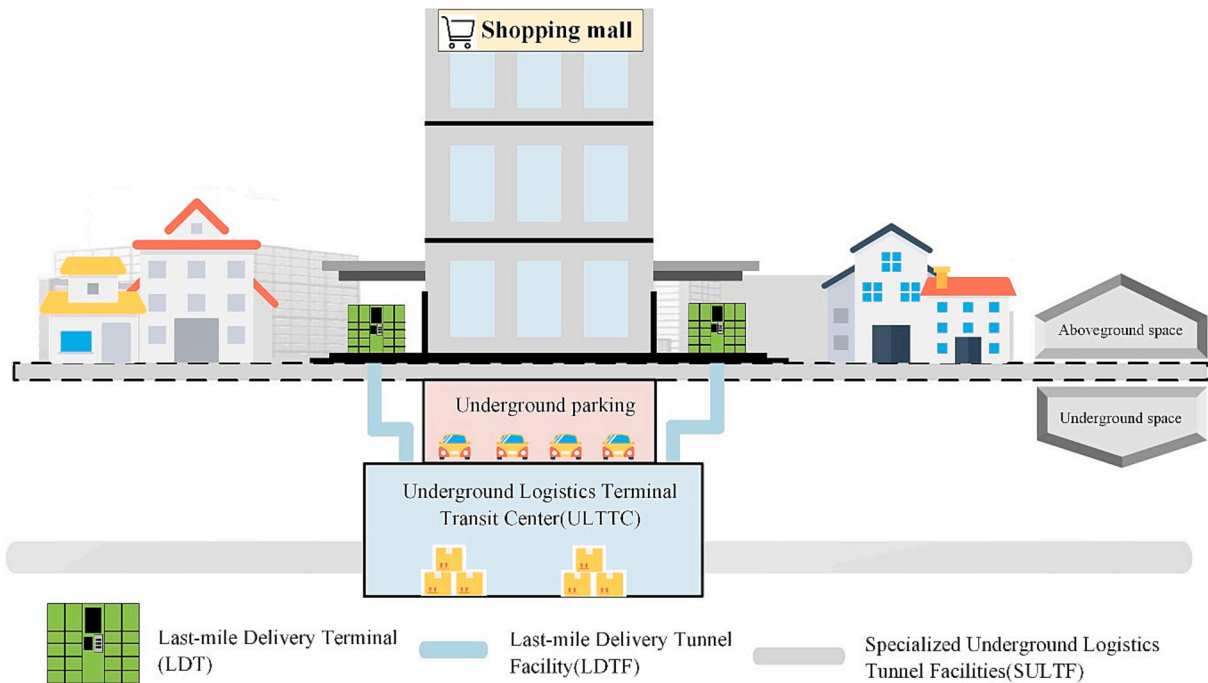


Fig. 6. Scheme of the “self-pickup + shopping malls” mode.

parcels from the designated LDT shopping mall and can send express parcels.

3.2. Design of the “home-entry +” mode

The “home-entry+” mode is suitable for two scenarios: office buildings and communities, which are discussed in detail below.

3.2.1. “Home-entry + office buildings” mode

The “home-entry + office building” mode involves the transportation of express parcels from ULTTCs, through LDTFs, to LDTs in office buildings. Customers do not even need to leave the floor to complete the delivery. In this study, we constructed two case studies of the “home-entry + office building” model, as shown in Fig. 7, where (a) the LDT is in the lobby on the first floor of the office building or (b) the LDTs are

distributed on each floor of the office building.

The “home-entry + office buildings” model works as follows: (1) After arriving at the ULTTC, the transported express parcels are unloaded and, using customer information, sorted according to the different LDTs in the office buildings; (2) Express parcels are delivered to the LDTs in the corresponding office buildings via LDTFs; (3) Customers pick up the express parcels at the designated LDT on each floor of the office buildings or in each area, and can deliver express parcels that need to be sent out or returned through that LDT.

3.2.2. “Home-entry + communities” mode

The “home-entry + communities” mode is for residential areas within which the LDT building is located. Express parcels are transported from the ULTTC, through the SULTF, to the LDT. The customer does not need to leave the floor for the delivery to be completed. For this

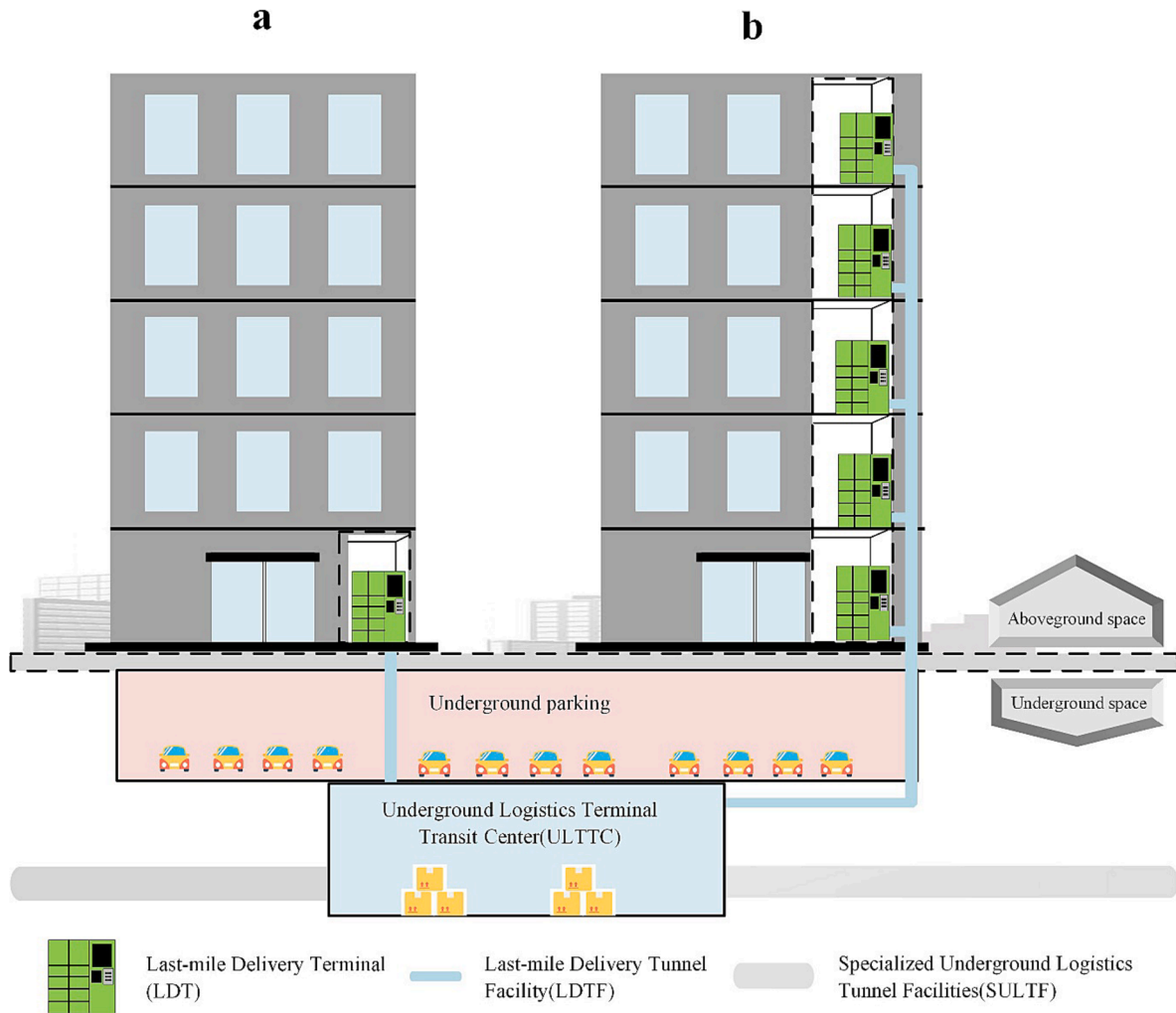


Fig. 7. Schematic design of the “home-entry + office buildings” mode: (a) the LDT is in the lobby on the first floor of the office building; (b) the LDT is distributed on each floor of the office building.

purpose, in this study, we constructed three scenarios for the “home-entry + communities” model, as shown in Fig. 8: (a) a scenario in which the LDT is within the first floor of a community unit building; (b) a scenario in which the LDT is in the common area of each floor of the unit building; and (c) a scenario in which the LDT is in each occupant’s doorway.

The “home-entry + communities” model works as follows: (1) After arriving at the ULTTC, the transported express parcels are unloaded and sorted according to different LDTs in the residential area using waybill customer information; (2) Express parcels are transported to the corresponding LDTs in the residential area via LDTFs; (3) Customers receive the parcels at their homes or pick them up in the public area of each floor or at the LDT on the first floor of each building. They can also drop off parcels that need to be sent or returned through this LDT.

4. Case study

4.1. Information and data of study area

The study area of this research was a city sub-center in China. The ULTTCs in the study area were established based on a service range of approximately 1 km from the LDT corresponding to the MFSL and ULTTC set up in this study. The study area is 14.3 km² and there is 1 MFSL in the study area, so 5 ULTTCs were set up so that their last-mile distribution terminal service area could include the entire study area.1

MFSL and 5 ULTTCs (a, b, c, d, e, and f in Fig. 9) were set up in the study area, as well as the service ranges of each LDT (Fig. 9). In this study, the LDTs corresponding to the MFSL and the five ULTTCs represented different scenarios, and the service scope of all LDTs covered the entire study area as follows: (a) the LDT at the MFSL is a “self-pickup + metro stations” model; (b) the LDT at the ULTTC is a “self-pickup + shopping malls” model; (c) the LDT at the ULTTC is a “home-entry + communities” model; (d) the LDT at the ULTTC is a “home-entry + office building & communities” model; (e) the LDT at the ULTTC is a “self-pickup + communities” model; and (f) the LDT at the ULTTC is a “self-pickup + communities” model.

The estimated number of express parcels delivered in the last-mile of the study area is approximately 363,000 per day, and the volume of outgoing (reverse logistics) express parcels is approximately 71,000 per day (Beijing General Municipal Engineering Design & Research Institute Co., Ltd, Army Engineering University of PLA, 2019; Guo et al., 2021). According to the research of the courier company, the average weight of express parcels = 5 kg / piece, which converts to an average volume of courier parcels = equaling 0.03 m³ / piece (31 × 31 × 31 cm cube) (Chen et al., 2021b; Chen et al., 2022). In this study, the dimensions of the pallets of express parcels transported in the LDTFs were 1.2 × 1 × 1.5 m; thus, when combined with the estimated volume of the express parcels, a pallet can carry approximately 60 express parcels at a time (Chen et al., 2021b; Chen et al., 2022).

Within the study area, the MFSL (a) is connected to the ULTTCs (b, c,



Fig. 8. Schematic design of the “home-entry + communities” mode: (a) a scenario in which the LDT is only within the first floor of a community unit building; (b) a scenario in which the LDT is in the common area of each floor of the unit building; and (c) a scenario in which the LDT is in the doorway of each occupant of the unit building.

d, e, f) by SULTFs, as shown in Fig. 10. There are many SULTF structures and, at present, the commonly used SULTF structures are: star, tree, ring, and mesh structures (Hu et al., 2023). Considering the area of the study area, the volume of express parcels, the number of ULTTCs, and the distance between the ULTTCs, the SULTFs in the study area were constructed to form a ring structure (Fig. 10). After all express parcels in the study area are transported via the SL to the MFSL (a), they are sorted within the MFSL (a) and transported to the rest of the ULTTCs via the constructed SULTFs. When there are express parcels that need to be sent out, each ULTTC transports the express parcels along SULTFs to the MFSL (a), where they are uniformly transported away from the study area by the SL.

4.2. Parameter settings

The parameters of the end-distribution pipeline facilities and end-distribution terminals were set for different scenarios in the study area as follows: (1) The “self-pickup + metro station” mode in the metro station has four exits, each having an LDT. Each LDT has a two-way circular terminal distribution pipeline to deliver express parcels, and each pipeline has four pallets working simultaneously. (2) The “self-pickup + shopping mall” mode comprises two exits in the shopping mall and one LDT at each exit, each LDT has one bi-directional circulating terminal delivery pipeline to deliver express parcels, and each pipeline has four pallets working simultaneously. (3) The “self-pickup + communities” mode comprises only one LDT in the residential area.

Considering that an LDT has to serve the whole residential area, the LDT was set up with two two-way circulating last-mile delivery pipelines to deliver express parcels, and each pipeline has eight pallets working simultaneously. (4) Both the “home-entry + communities” mode and the “home-entry + office building” mode have one LDT for each floor and, after investigation, there are 55 residential buildings in the last-mile delivery area at the ULTTC (c), and 96 residential buildings and four office buildings in the last-mile delivery area at the ULTTC (d), therefore there is only one bi-directional circulating last-mile delivery pipeline connecting all the LDTs in each building. (5) Express parcels in this study were transported via SLs that are common passenger transportation structures. A survey in the study area revealed that the operating hours of the metro passenger transportation were from 5:00 to 24:00, and the morning and evening peak hours were from 6:00 to 9:00 and from 17:00 to 20:00. Considering that the morning and evening peak passenger flows are large, in order not to affect the metro’s normal passenger transportation, the last-mile delivery times are typically too early or too late, thus inconvenient for customers to collect or deliver express parcels. Therefore, in this study, the metro transportation of express parcels was set from 9:30 to 16:30. The frequency of express parcel transportation through the metro to the study area was every half hour, and the specific metro freight moment diagram is shown in Fig. 11. (6) A total of 15 trips a day were made and each shipment was the same amount when the express parcel arrived at the MFSL in the study area immediately after the next stage of urban underground logistics transportation, while the reverse logistics of the express parcels away from

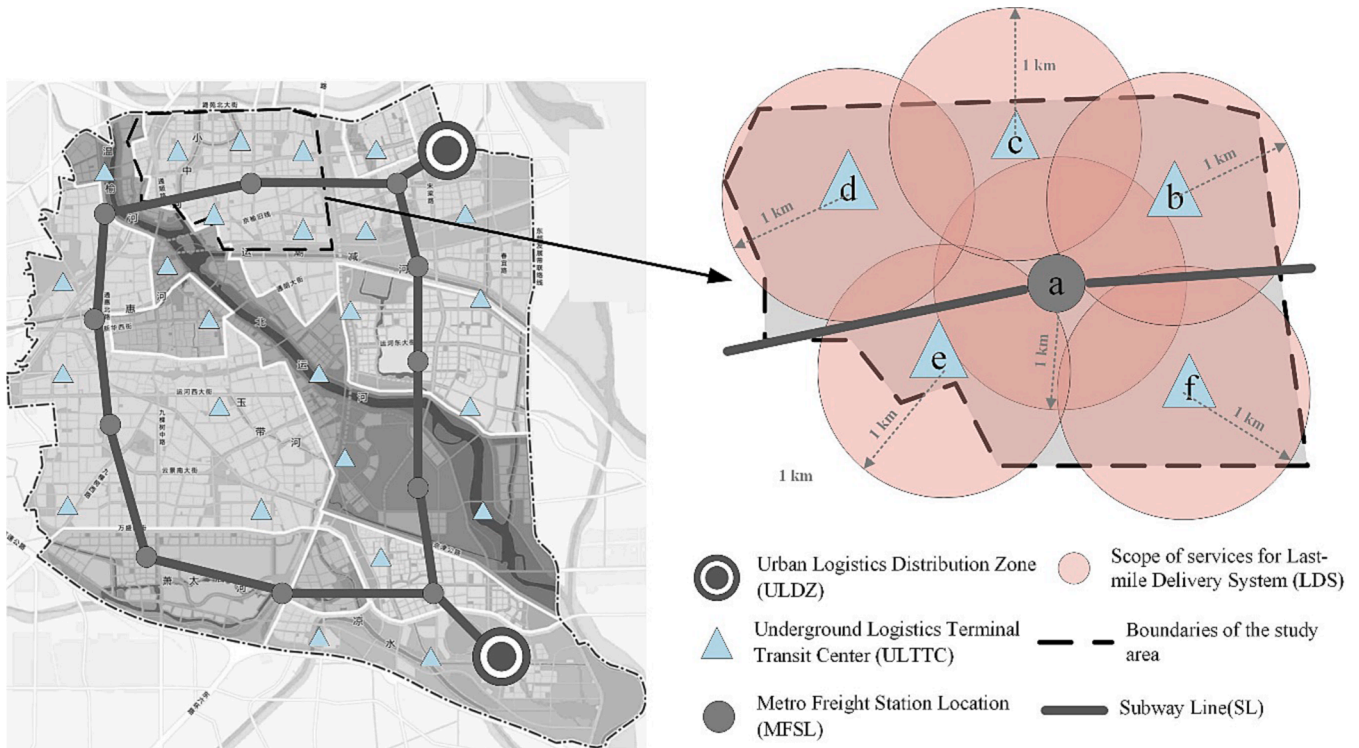


Fig. 9. Spatial location map of the case study area.

the study area were the same amount each time. (7) This study set up a freight metro, with a carriage that was 20 m long, 3 m wide, 3.5 m high, and four carriages were used to form a train for subway line transportation; it can be seen from Fig. 11 that 15 freight trains per day stop at MFSL in the study area. According to the calculation of 90 % of the maximum volume of each freight carriage, the capacity of freight transported to MFSL in the study area is $11,340 \text{ m}^3$ ($20 \times 3 \times 3.5 \times 4 \times 15 \times 90\%$); the theoretical number of express parcels delivered to MFSL in the study area was calculated to be 378,000 per day based on an average volume of 0.03 m^3 /piece (Chen et al., 2021b; Chen et al., 2022), which is greater than the estimated daily number of express parcels (363,000 per day) for LDS in the study area, and the freight capacity of the subway lines to satisfy the demand for last-mile delivery freights. (8) As the LDTF transports express parcels in the MFSL to each ULTTC over time, it is assumed that the arrival distribution of express parcels in each scenario is consistent with the times of the metro transporting express parcels of fifth setting.

In addition, based on the above parameter settings, the parameters related to the LDTF (including pallet carriers and corresponding transportation lanes) were set as follows: (1) Pallets in the LDTF moved at a constant speed of 1 m/s and carried the same number of express parcels each time (60 pieces). (2) The time for loading and unloading express parcels in the ULTTC and the LDT was fixed at 30 s, which did not change with changes in the number of express parcels that were loaded and unloaded. (3) All MFSLs, ULTTCs, and LDTs had enough space to store express parcels. (4) It was assumed that the average length of the pipeline from the MFSL to each LDT in the scenario of the “self-pickup + metro station” mode is 100 m; the average length of the pipeline from the ULTTC to each LDT in the “self-pickup + shopping malls” and “self-pickup + communities” scenarios is 50 m; and the average length of the pipeline from the ULTTC to the 1st floor of each building in the “home-entry +” mode is 500 m, and the communities and office buildings are all 10-story buildings, with a height of 3 m for each story. (5) It was assumed that “forward logistics” and “reverse logistics” did not occur at the same time in the LDTFs in each scenario.

4.3. Calculation formula for last-mile delivery system

Reverse logistics will not be carried out during the return process after the pallet carriers have completed forward logistics in the scenarios of “self-pickup + metro station”, “self-pickup + communities” and “self-pickup + shopping malls.” Owing to the long length of the two-way circulating last-mile delivery pipelines of “home-entry + office buildings” and “home-entry + communities,” it is assumed that reverse logistics will be carried out during the return process after the pallet carriers have completed forward logistics, “in order to save the time of LDS and improve the utilization efficiency of pallet carriers. The formula for calculating the freight volume per subway shift for forward/reverse logistics for each scenario is as follows:

$$q = Q/S \quad (1)$$

where, q denotes the amount of forward/reverse logistics shipments per subway shift for each scenario, in pieces/day; Q denotes the total number of forward/reverse logistics express parcels in one day for each scenario, in pieces/day; and S denotes the number of subway shifts in one day in the study area, in shifts/day.

The total time forward/reverse logistics of last-mile delivery for each scenario of the “self-pickup +” mode is calculated as follows:

$$T_{\text{self-pickup}}^{\text{total}} = (2L/v + t_1 + t_2) \times Q/(60 \times n \times p) \quad (2)$$

where, $T_{\text{self-pickup}}^{\text{total}}$ denotes the total time forward/reverse logistics of last-mile delivery for each scenario of the “self-pickup +”, in minutes; L denotes the length of the bi-directional circulating terminal delivery pipeline for each scenario, in m; v denotes the velocity of the pallet carrier in the pipe, in m/s; t_1 denotes the time of loading and unloading express parcels of the pallet carrier at the terminal of MFSL, in s; t_2 denotes the time of loading and unloading express parcels of the pallet carrier at the terminal of LDS, in s; n denotes the number of express parcels carried on the pallet carrier at one time, in pieces; p denotes the number of pallet carriers for each scenario.

The total time forward/reverse logistics of last-mile delivery for each

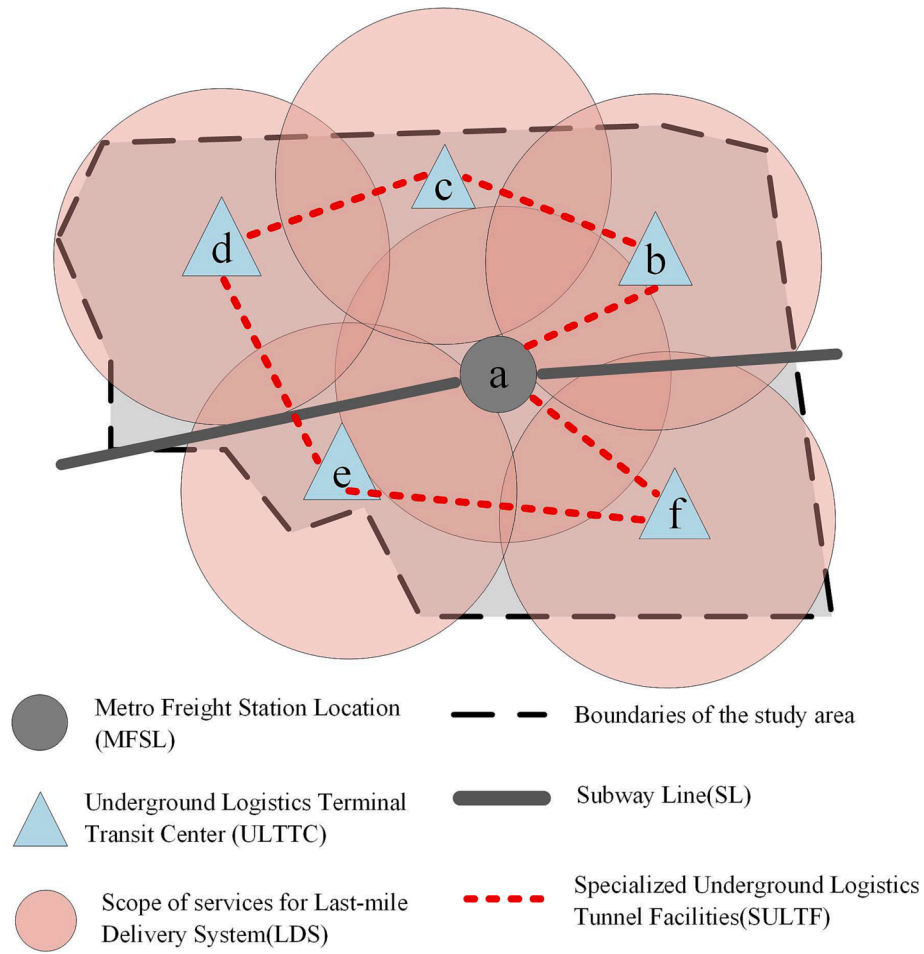


Fig. 10. Schematic diagram of MFSL connecting to other ULTTCs in the study area.

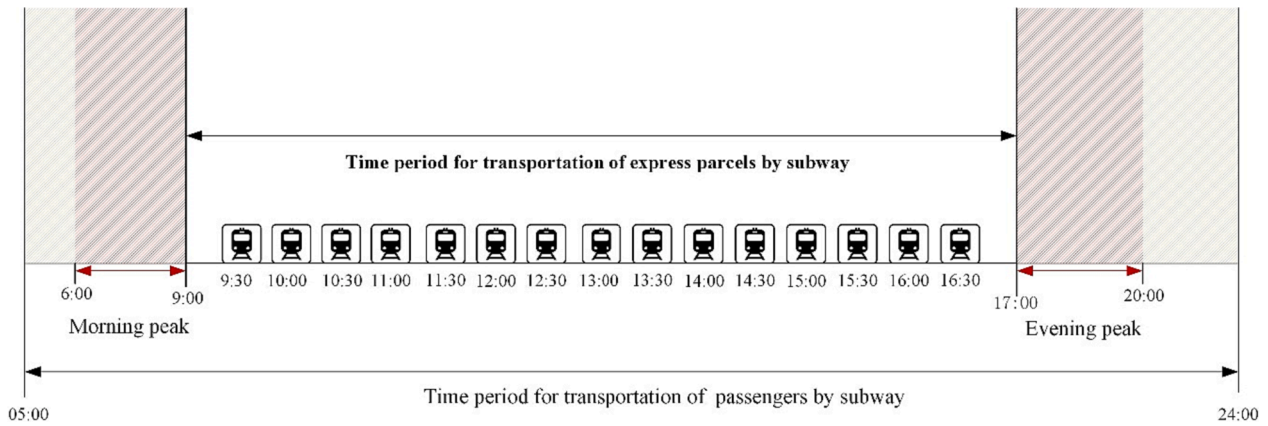


Fig. 11. Diagram of specific subway freight moment.

scenario of the “home-entry +” mode is calculated as follows:

$$T_{\text{home-entry}}^{\text{total}} = (L/v + t_1 + t_2) \times Q / (60 \times n \times p) \quad (3)$$

where, $T_{\text{home-entry}}^{\text{total}}$ denotes the total time forward/reverse logistics of last-mile delivery for each scenario of the “home-entry +”, in minutes.

The time per metro shift of forward/reverse logistics of last-mile delivery for each scenario of the “self-pickup +” mode is calculated as follows:

$$T_{\text{self-pickup}}^{\text{per}} = (2L/v + t_1 + t_2) \times q / (60 \times n \times p) \quad (4)$$

where, $T_{\text{self-pickup}}^{\text{per}}$ denotes time of per metro shift forward/reverse logistics of last-mile delivery for each scenario of the “self-pickup +” mode, in minutes.

The time per metro shift of forward/reverse logistics of last-mile delivery for each scenario of the “home-entry +” mode is calculated as follows:

$$T_{\text{home-entry}}^{\text{per}} = (L/v + t_1 + t_2) \times q / (60 \times n \times p) \quad (5)$$

where, $T_{\text{home-entry}}^{\text{per}}$ denotes time per metro shift of forward/reverse logistics of last-mile delivery for each scenario of the “home-entry +” mode, in minutes.

The average delivery time for a single express parcel (seconds) for each scenario of the “self-pickup +” mode is calculated as follows:

$$T_{\text{self-pickup}}^{\text{average}} = 60T_{\text{self-pickup}}^{\text{total}} / Q \quad (6)$$

The average delivery time for a single express parcel (seconds) for each scenario of the “home-entry +” mode is calculated as follows:

$$T_{\text{home-entry}}^{\text{average}} = 60T_{\text{home-entry}}^{\text{total}} / Q \quad (7)$$

4.4. Results

According to the above parameter settings for the study area, combined with the number of express parcels transported by the MFSs and ULTTs per metro shift of forward and reverse logistics, the transport parameters for LDS in each scenario were calculated, as shown in Table 1. Further, graphs of the forward and reverse LDT times for express parcel volumes in the five scenarios were plotted accordingly, as in Fig. 12.

As shown in Table 1 and Fig. 12, (1) overall, comparing forward and reverse logistics in each scenario, when the number of pallets is the same, the longer the length of the two-way circular last-mile delivery pipeline, the longer the average last-mile delivery time; (2) For the three “self-pickup +” modes, the express parcel number in the “self-pickup +

Table 1
The transport parameters for LDS in each scenario.

Parameters	(a)	(b)	(c)	(d)	(e)	(f)
Number of pallets	16	8	55	100	16	16
Length of bi-directional circulating last-mile delivery pipeline (m)	100	50	530	530	50	50
Total number of forward logistics express parcels in one day (1,000 pieces)	40	35	40	72	96	80
Total number of reverse logistics express parcels in one day (1,000 pieces)	6	10	8	20	17	10
Volume per metro shift of forward logistics (1,000 pieces)	2.7	2.3	2.7	4.8	6.4	5.3
Volume per metro shift of reverse logistics (pieces)	533	400	533	867	1334	1067
Total time of forward logistics of last-mile delivery (minutes)	180.6	191.3	173.7	172.0	266.7	222.3
Time of per metro shift of forward logistics of last-mile delivery (minutes)	12.04	12.75	11.58	11.47	17.78	14.82
Total time of reverse logistics of last-mile delivery (minutes)	27.1	55.6	34.7	47.8	47.2	27.8
Time of per metro shift of reverse logistics of last-mile delivery (minutes)	1.81	3.71	2.31	3.19	3.15	1.85
Average delivery time for a single express parcel (seconds)	0.271	0.328	0.261	0.143	0.167	0.167

Explanation of abbreviations in the table: (a) is a “self-pickup + metro stations” mode; LDT at ULTT (b) is a “self-pickup + shopping malls” mode; LDT at ULTT (c) is a “home-entry + communities” mode; LDT at ULTT (d) is a “home-entry + office buildings & communities” mode; LDT at ULTT (e) is a “self-pickup + communities” mode; LDT at ULTT (f) is a “self-pickup + communities” mode.

shopping malls” scenario in the forward logistics was smaller than that of the “self-pickup + metro stations” scenario, but the time for last-mile delivery in the forward logistics in the “self-pickup + shopping malls” scenario was greater than that in the “self-pickup + metro stations” scenario. This is because the number of pallets in the LDTs in the “pick-up + shopping malls” scenario was less than that in the “self-pickup + metro stations” scenario, resulting in a longer average time for express parcels in last-mile delivery. (3) For the two “home-entry+” modes, the number of forward logistics express parcels in the “home-entry + office buildings & communities” scenario was larger than that in the “home-entry + communities” scenario. However, the forward logistics last-mile delivery time in the “home-entry + office buildings & communities” scenario was smaller than that in the “home-entry + communities” scenario. This is because the number of pallets in the LDTs of the “home-entry + office buildings & communities” scenario is higher than that in the “home-entry + communities” scenario, resulting in a shorter average last-mile delivery time for express parcels.

5. Discussion and analysis

5.1. Comparison of delivery efficiency with above ground parcel distribution

Currently, last-mile delivery also relies on couriers in cities to carry out the final stage of above-ground parcel distribution. This requires many couriers to work long hours, resulting in traffic congestion, environmental pollution, space constraints, and other city problems. We researched courier companies and couriers to obtain relevant data for the city where the study area is located: the current efficiency of last-mile delivery is 300 pieces/person/day, with a daily working time of 10 to 12 h. The number of couriers and delivery time required for the current last-mile delivery strategy were calculated based on the current last-mile delivery efficiency and compared with those of the last-mile delivery strategy constructed in this study, as shown in Table 2. In particular, we set up each scenario to require 10 workers for the control and management of LDS, based on the number of workers required for automated freight handling on the ground as a reference basis.

As shown in Table 2, for the same amount of express parcels, the current LDS requires 1550 couriers, whereas the LDS only requires minimal management personnel, and greatly reduces human resources wastage and human influence on last-mile delivery time caused by time delays. Regarding distribution time, at the present stage, last-mile delivery distribution takes 10 to 12 h, and efficiency is low, failing to meet future customer demand for highly efficient last-mile delivery. In contrast, distribution time in the LDS constructed in this study was only 3.4 to 5.2 h, which is greatly reduced compared to the time of the current LDS.

5.2. Analysis of emission reductions for surface transportation

CO₂ emissions during the operation of the LDS constructed in this study was compared with the surface transportation for LDS. We determined the dimensions, load capacity, and energy consumption through a survey for the electric tricycles used for last-mile surface transportation delivery at the current stage (<https://item.jd.com/10074526060933.html> accessed on February 10, 2024). For the relevant operating parameters of the pallet carrier in this study, we refer to the relevant energy consumption parameters of the heavy-duty pallet four-way shuttle (<http://www.up-edge.com/nd.jsp?id=42&fromMid=425> accessed on February 10, 2024) designed by Prologic Smartcore Technology (Hubei) Co. Ltd. which has a similar size and load capacity as the pallet carrier in this study. The carbon emission factor for electricity is 0.9419kgCO₂/kWh (National Development and Reform Commission of the People's Republic of China, 2019). The year 2030 is taken as the year of LDS completion, and the average growth rate of express service

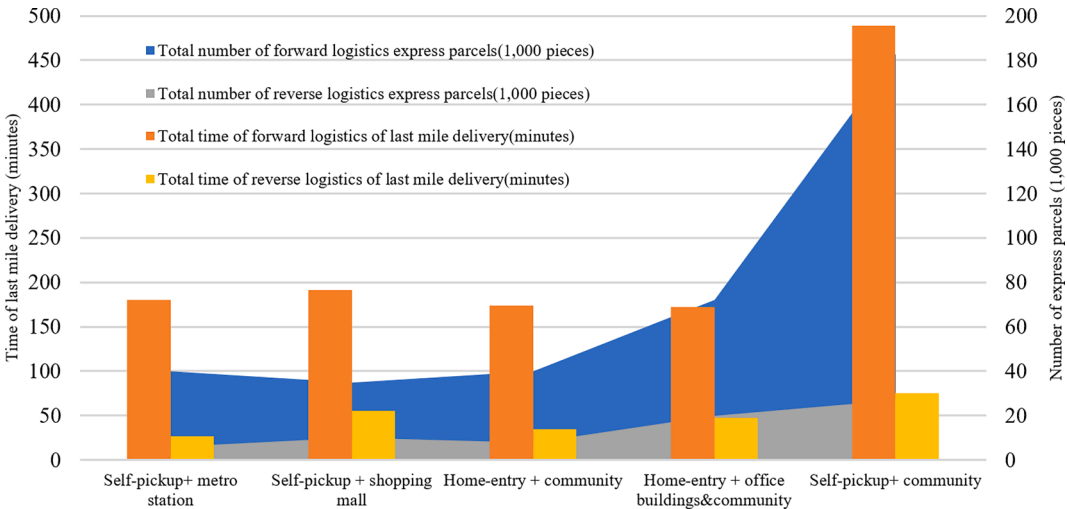


Fig. 12. Graphs of reverse LDT times for express parcel volumes in the five different scenarios.

Table 2
Comparison of the efficiency of last-mile delivery currently and this study.

Classifications	Total express parcels (1,000 pieces)	Number of couriers (persons)	Last-mile delivery time (hours)
The current LDS	434	1550	10 ~ 12
The LDS in this study	434	60	3.4 ~ 5.2

volume in Beijing from 2012 to 2020 is taken as the annual growth rate of express parcels for LDS in the study area. The carbon emissions from surface transportation for LDS and emissions reduction of LDS by using ULS in 2030–2050 are calculated as shown in Fig. 13. As can be seen in

Fig. 13, the use of ULS can reduce carbon emissions of surface transportation for LDS by 96 %, and the contribution of the use of ULS to the reduction of carbon emissions for surface transportation becomes more and more significant over time, with a reduction of 1,829 CO₂t per year by 2050.

5.3. Analysis of the advantages of LDS

In this study, we constructed a multi-scenario express parcel transportation and distribution model for the LDS of ULS and calculated the distribution time of last-mile delivery for each scenario by setting and assuming the conditions in the study area. The LDS of ULS constructed in this study includes both urban logistics transportation and last-mile

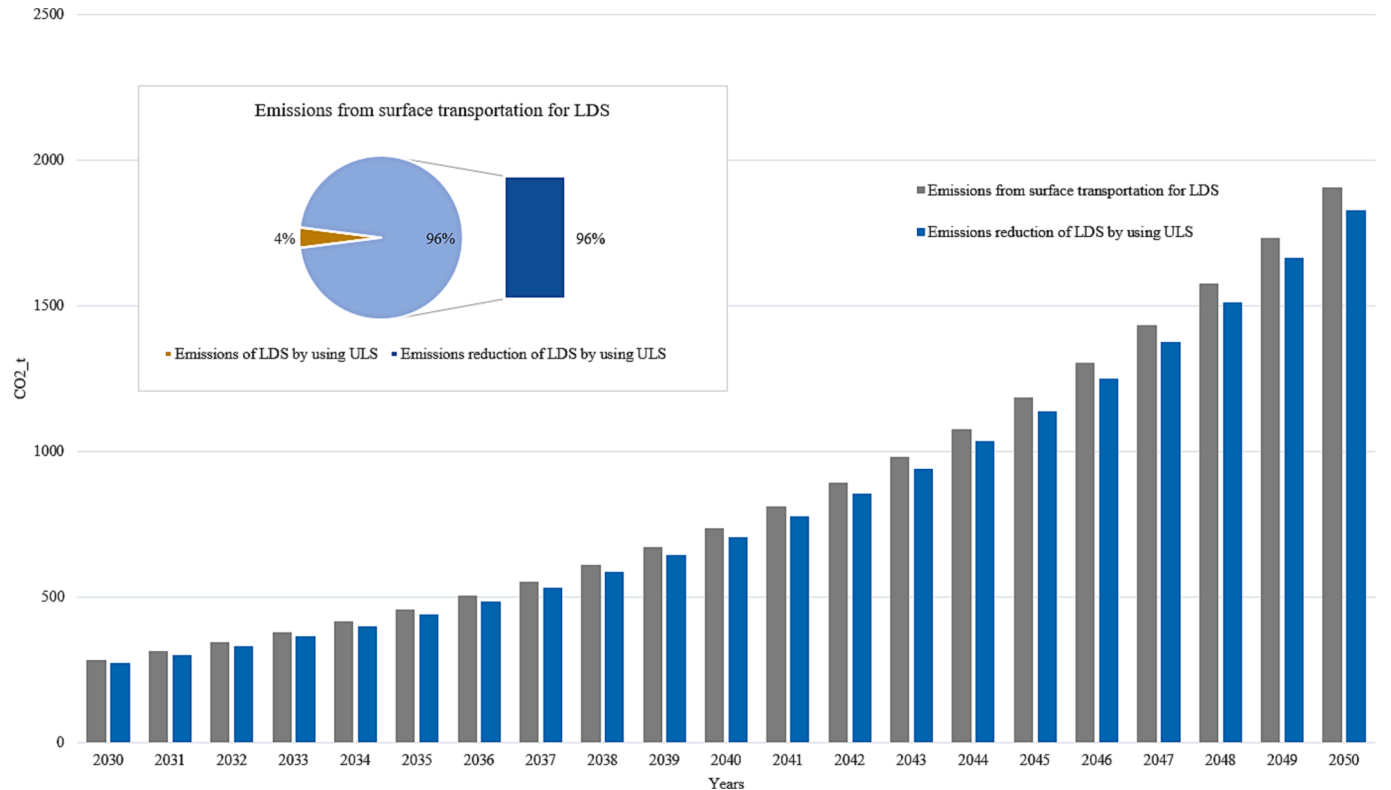


Fig. 13. Carbon emissions from surface transportation for LDS and emissions reduction of LDS by using ULS.

delivery of express parcels, which have significant advantages, and are described below from urban logistics, last-mile delivery, and customer perspectives.

- The urban logistics perspective: in this study, express parcels were transported using the remaining capacity of metro passenger transportation, thus greatly reducing ULS construction costs. In addition, the MFSL and ULTTC are built underground, which is conducive to solving the traffic problems in cities caused by logistics transportation, such as traffic congestion and accidents. Overall, the LDS of ULS constructed in this study saves urban ground space resources, reduces the emission of pollutant gases during transportation, and contributes to the protection of the urban environment.
- The last-mile delivery perspective: as the urban express parcel transportation system in this study is located underground without the influence of above ground transportation, and since the last-mile delivery process requires the SLs, SULTFs, and LDTFs for completion, the efficiencies of both logistics transportation and last-mile delivery are greatly improved. This advantage is also confirmed by comparisons with the characteristics of the current LDS, as the last-mile delivery process in this study does not require couriers (only a minimal number of management personnel), dramatically saving social human resources.
- The customer perspective: as the delivery stage of express parcel distribution, last-mile delivery has a great impact on production and the lives of customers. According to the actual needs of customers in the communities, shopping malls, and office buildings, the five different scenarios and modes of LDS constructed in this study resulted in shorter last-mile delivery times, enabling customers to complete the delivery more conveniently and quickly. The last-mile delivery modes without courier participation also make the delivery process more secure and reliable.

6. Conclusions

A case study in a city sub-center showed that both the forward logistics and reverse logistics of last-mile delivery can be accomplished across all five scenarios specified in this study, with the same pallet capacity and movement speed in the LDTFs. The efficiency of last-mile delivery in our LDS (i.e., the average distribution time of a single express parcel) is closely related to the length of the bi-directional circular end distribution pipeline in the LDTFs and number of pallets. Specifically, for the “self-pickup +” mode with the same number of pallets, the longer the length of the bi-directional circular end distribution pipeline, the longer the time required for the forward and reverse logistics for last-mile delivery, and the longer the average distribution time of a single express parcel. For the “self-pickup +” and “home-entry +” modes, the length of the two-way circular distribution pipeline at the final stage is the same. The greater the number of pallets, the shorter the duration of forward and reverse logistics for last-mile delivery and the shorter the average distribution time of a single express parcel. In addition, the use of ULS can reduce carbon emissions of surface transportation for LDS by 96 %, and the contribution of the use of ULS to the reduction of carbon emissions for surface transportation becomes more and more significant over time, with a reduction of 1,829 CO₂t per year by 2050.

However, it should be noted that this LDS is a new ULS model, and the model and scheme proposed in this paper may have some limitations due to the lack of engineering practice. The following limitations and assumptions of this study should be noted. First, although we constructed different modes of LDSs, we only considered feasibility from the last-mile delivery operation perspective and not from an economic perspective. For example, this LDS would require transformation of existing metro stations, construction of ULTTCs in the underground spaces of the city, and construction of LDTFs of different lengths to connect ULTTCs and different LDTs, in the “self-pickup+” and “home-entry +” modes, which all require higher economic costs. Second, the

assumptions made in the case study are relatively simple and idealized; for example, the assumption that pallets of express parcels will move at a constant speed without considering acceleration and deceleration, the loading and unloading times being set to a fixed value, the number of floors being set to 10 floors in the “home-entry +” mode, etc. These helped verify the feasibility of this study’s LDS from an operational perspective; however, further research for specific implementation at operation levels is required.

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CRediT authorship contribution statement

Lingxiang Wei: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft. **Yongfan Chen:** Conceptualization, Data curation, Methodology, Validation, Visualization. **Dongjun Guo:** Funding acquisition, Project administration, Validation, Visualization. **Junyuan Ji:** Investigation, Methodology, Validation, Visualization. **Zhilong Chen:** Investigation, Supervision, Writing – review & editing. **Changchun Zhuo:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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References

- Aized, T., Srai, J.S., 2014. Hierarchical modelling of last mile logistic distribution system. *Int. J. Adv. Manuf. Technol.* 70, 1053–1061. <https://doi.org/10.1007/s00170-013-5349-3>.
- Beijing General Municipal Engineering Design & Research Institute Co., Ltd, Army Engineering University of PLA., 2019. Interim Report for Planning of Underground Logistics System for the Sub-centers of Beijing. Beijing.
- Bobylev, N., 2009. Mainstreaming sustainable development into a city’s master plan: a case of urban underground space use. *Land Use Policy* 26 (4), 1128–1137. <https://doi.org/10.1016/j.landusepol.2009.02.003>.
- Bobylev, N., Sterling, R., 2016. Urban underground space: a growing imperative. Perspectives and current research in planning and design for underground space use. *Tunnelling Undergr. Space Technol.* 55, 1–4. <https://doi.org/10.1016/j.tust.2016.02.022>.
- Broere, W., 2016. Urban underground space: solving the problems of today’s cities. *Tunnelling Undergr. Space Technol.* 55, 245–248. <https://doi.org/10.1016/j.tust.2015.11.012>.
- Chen, C., Demir, E., Huang, Y., Qiu, R., 2021a. The adoption of self-driving delivery robots in last mile logistics. *Transp. Res. E Logist. Transp. Rev.* 146, 102214 <https://doi.org/10.1016/j.tre.2020.102214>.
- Chen, Z., Dong, J., Ren, R., 2017. Urban underground logistics system in China: opportunities or challenges? *Undergr. Space* 2, 195–208. <https://doi.org/10.1016/j.undsp.2017.08.002>.
- Chen, Y., Guo, D., Chen, Z., Fan, Y., Li, X., 2018b. Using a multi-objective programming model to validate feasibility of an underground freight transportation system for the yangshan port in Shanghai. *Tunnelling Undergr. Space Technol.* 81, 463–471. <https://doi.org/10.1016/j.tust.2018.07.012>.

- Chen, D., Ignatius, J., Sun, D., Goh, M., Zhan, S., 2018a. Impact of congestion pricing schemes on emissions and temporal shift of freight transport. *Transp. Res. E Logist. Transp. Rev.* 118, 77–105. <https://doi.org/10.1016/j.tre.2018.07.006>.
- Chen, Y., Liu, Y., Guo, D., Chen, Z., Li, X., 2020. Freight transit assignments for an integrated network of road transportation and underground logistics systems. *J. Pipeline Syst. Eng. Pract.* 11 (2) [https://doi.org/10.1061/\(asce\)ps.1949-1204.0000434](https://doi.org/10.1061/(asce)ps.1949-1204.0000434).
- Chen, Y., Guo, D., Tao, X., Chen, Z., Zhang, W., 2021b. Forecast analysis of freight demand volume in the urban underground logistics system. *Chinese J. Underground Space and Eng.* 17 (06), 1687–1694 in Chinese.
- Chen, Y., Guo, D., Zhang, Y., Zhao, Z., 2022. Analysis of last-mile distribution efficiency of urban underground logistics based on AnyLogic. *J. Army Eng. Univ. PLA* 1 (05), 74–79 in Chinese.
- Chen, Z., Hu, W., Xu, Y., Dong, J., Yang, K., Ren, R., 2023. Exploring decision-making mechanisms for the metro-based underground logistics system network expansion: an example of Beijing. *Tunnelling Undergr. Space Technol.* 139, 105240 <https://doi.org/10.1016/j.tust.2023.105240>.
- Cui, J., Broere, W., Lin, D., 2021. Underground space utilisation for urban renewal. *Tunnelling Undergr. Space Technol.* 108, 103726 <https://doi.org/10.1016/j.tust.2020.103726>.
- Di, Z., Yang, L., Shi, J., Zhou, H., Yang, K., Gao, Z., 2022. Joint optimization of carriage arrangement and flow control in a metro-based underground logistics system. *Transp. Res. B Methodol.* 159, 1–23. <https://doi.org/10.1016/j.trb.2022.02.014>.
- Ding, A., Zhang, Y., Zhu, L., Du, Y., Ma, L., 2020. Recognition method research on rough handling of express parcels based on acceleration features and CNN. *Measurement* 163, 107942. <https://doi.org/10.1016/j.measurement.2020.107942>.
- Esper, T.L., Jensen, T.D., Turnipseed, F.L., Burton, S., 2003. The last mile: an examination of effects of online retail delivery strategies on consumers. *J. Bus. Logist.* 24, 177–203. <https://doi.org/10.1002/j.2158-1592.2003.tb00051.x>.
- Fan, Y., Liang, C., Hu, X., Li, Y., 2020. Planning connections between underground logistics system and container ports. *Comput. Ind. Eng.* 139, 106199 <https://doi.org/10.1016/j.cie.2019.106199>.
- Gevaers, R., Van de Voorde, E., Vanelander, T., 2014. Cost modelling and simulation of last-mile characteristics in an innovative B2C supply chain environment with implications on urban areas and cities. *Procedia Soc. Behav. Sci.* 125, 398–411. <https://doi.org/10.1016/j.sbspro.2014.01.1483>.
- Guo, D., Chen, Y., Yang, J., Tan, Y.H., Zhang, C., Chen, Z., 2021. Planning and application of underground logistics systems in new cities and districts in China. *Tunnelling Undergr. Space Technol.* 113, 103947 <https://doi.org/10.1016/j.tust.2021.103947>.
- Guo, X., Luján Jaramillo, Y.J., Bloemhof-Ruwaard, J., Claassen, G.D.H., 2019. On integrating crowdsourced delivery in last-mile logistics: a simulation study to quantify its feasibility. *J. Clean. Prod.* 241, 118365 <https://doi.org/10.1016/j.jclepro.2019.118365>.
- Hai, D., Xu, J., Duan, Z., Chen, C., 2020. Effects of underground logistics system on urban freight traffic: a case study in Shanghai. *China. J. Cleaner Prod.* 260, 121019 <https://doi.org/10.1016/j.jclepro.2020.121019>.
- He, M., Sun, L., Zeng, X., Liu, W., Tao, S., 2020. Node layout plans for urban underground logistics systems based on heuristic bat algorithm. *Comput. Commun.* 154, 465–480. <https://doi.org/10.1016/j.comcom.2020.02.075>.
- Hu, W., Dong, J., Hwang, B.G., Ren, R., Chen, Z., 2021. A preliminary prototyping approach for emerging metro-based underground logistics systems: operation mechanism and facility layout. *Int. J. Prod. Res.* 59 (24), 7516–7536. <https://doi.org/10.1080/00207543.2020.1844333>.
- Hu, W., Dong, J., Xu, N., 2022a. Multi-period planning of integrated underground logistics system network for automated construction-demolition-municipal waste collection and parcel delivery: a case study. *J. Clean. Prod.* 330, 129760 <https://doi.org/10.1016/j.jclepro.2021.129760>.
- Hu, W., Dong, J., Yuan, J., Ren, R., Chen, Z., Cheng, H., 2022b. Agent-based modeling approach for evaluating underground logistics system benefits and long-term development in megacities. *J. Manag. Sci. Eng.* 7 (2), 266–286. <https://doi.org/10.1016/j.jmse.2021.10.002>.
- Hu, W., Dong, J., Yang, K., Hwang, B.G., Ren, R., Chen, Z., 2023. Modeling real-time operations of metro-based urban underground logistics system network: a discrete event simulation approach. *Tunnelling Undergr. Space Technol.* 132, 104896 <https://doi.org/10.1016/j.tust.2022.104896>.
- Jiang, Y., Song, G., Zhang, H., 2023. Material identification and heavy metal characteristics of plastic packaging bags used in chinese express delivery. *Front. Environ. Sci.* 11, 1253108. <https://doi.org/10.3389/fenvs.2023.1253108>.
- Kim, M., Kwon, Y., Kim, J., Kim, Y., 2022. Image classification of parcel boxes under the underground logistics system using CNN MobileNet. *Appl. Sci.* 12, 3337. <https://doi.org/10.3390/ap12073337>.
- Kummer, S., Hribernik, M., Herold, D.M., et al., 2020. The impact of courier-, express- and parcel (CEP) service providers on urban road traffic: the case of Vienna. *Transp. Res. Interdiscip. Perspect.* 9, 100278 <https://doi.org/10.1016/j.trip.2020.100278>.
- Liang, C., Hu, X., Shi, L., Fu, H., Xu, D., 2022. Joint dispatch of shipment equipment considering underground container logistics. *Comput. Ind. Eng.* 165, 107874 <https://doi.org/10.1016/j.cie.2021.107874>.
- Lin, D., Broere, W., Cui, J., 2022. Underground space utilisation and new town development: experiences, lessons and implications. *Tunnelling Undergr. Space Technol.* 119, 104204. <https://doi.org/10.1016/j.tust.2021.104204>.
- Ma, M., Zhang, F., Liu, W., Dixit, V., 2022. A game theoretical analysis of metro-integrated city logistics systems. *Transp. Res. B Methodol.* 156, 14–27. <https://doi.org/10.1016/j.trb.2021.12.005>.
- Marcucci, E., Le Pira, M., Gatta, V., Inturri, G., Ignaccolo, M., Pluchino, A., 2017. Simulating participatory urban freight transport policy-making: accounting for heterogeneous stakeholders' preferences and interaction effects. *Transp. Res. E Logist. Transp. Rev.* 103, 69–86. <https://doi.org/10.1016/j.tre.2017.04.006>.
- Mo, P., Yao, Y., D'Ariano, A., Liu, Z., 2023. The vehicle routing problem with underground logistics: formulation and algorithm. *Transportation Research Part e: Logistics and Transportation Review* 179, 103286. <https://doi.org/10.1016/j.tre.2023.103286>.
- National Development and Reform Commission of the People's Republic of China, 2019. China regional grid baseline emission factors for FY2019 emission reduction programs. Beijing, China (in Chinese).
- Nuzzolo, A., Comi, A., Ibeas, A., Moura, J.L., 2016. Urban freight transport and city logistics policies: indications from Rome, Barcelona, and Santander. *Int. J. Sustain. Transp.* 10, 552–566. <https://doi.org/10.1080/15568318.2015.1014778>.
- Ouyang, Z., Leung, E.K.H., Huang, G.Q., 2023. Community logistics and dynamic community partitioning: a new approach for solving e-commerce last mile delivery. *Eur. J. Oper. Res.* 307, 140–156. <https://doi.org/10.1016/j.ejor.2022.08.029>.
- Pan, X., Dong, J., Ren, R., Chen, Y., Sun, B., Chen, Z., 2023. Monetary evaluation of the external benefits of urban underground logistics system: a case study of Beijing. *Tunnelling Undergr. Space Technol.* 136, 105094 <https://doi.org/10.1016/j.tust.2023.105094>.
- Pan, Y., Liang, C., Dong, L., 2019. A two-stage model for an urban underground container transportation plan problem. *Comput. Ind. Eng.* 138, 106113 <https://doi.org/10.1016/j.cie.2019.106113>.
- Patella, S.M., Grazieschi, G., Gatta, V., Marcucci, E., Carrese, S., 2020. The adoption of green vehicles in last mile logistics: a systematic review. *Sustainability* 13, 6. <https://doi.org/10.3390/su13010006>.
- Perboli, G., Rosano, M., Saint-Guillain, M., Rizzo, P., 2018. Simulation-optimisation framework for city logistics: an application on multimodal last-mile delivery. *IET Intell. Transp. Syst.* 12, 262–269. Portico. 10.1049/iet-its.2017.0357.
- Ranieri, L., Digiesi, S., Silvestri, B., Roccotelli, M., 2018. A review of last mile logistics innovations in an externalities cost reduction vision. *Sustainability* 10, 782. <https://doi.org/10.3390/su10030782>.
- Rezaeifar, F., Najafi, M., Kaushal, V., Huff, B., 2022. Development of a model to optimize the operations of an intermodal underground logistics transportation. *J. Pipeline Syst. Eng. Pract.* 13 (4), 04022039. [https://doi.org/10.1061/\(ASCE\)PS.1949-1204.0000678](https://doi.org/10.1061/(ASCE)PS.1949-1204.0000678).
- Shao, S., Xu, G., Li, M., Huang, G.Q., 2019. Synchronizing e-commerce city logistics with sliding time windows. *Transp. Res. E Logist. Transp. Rev.* 123, 17–28. <https://doi.org/10.1016/j.tre.2019.01.007>.
- State Post Bureau of The People's Republic of China, 2022. China express delivery development index report in 2022. accessed on February 10, 2024. <https://www.spb.gov.cn/gjywj/c100015/c100016/202301/c910dd57e739490ea6b0da58174ef826.shtml>.
- Vernimmen, B., Dullaert, W., Geens, E., Notteboom, T., T'Jollyn, B., Van Gilsen, W., Winkelmans, W., 2007. Underground logistics systems: a way to cope with growing internal container traffic in the port of Antwerp? *Transp. Plan. Technol.* 30, 391–416. <https://doi.org/10.1080/03081060701461832>.
- Villa, R., Monzón, A., 2021. A metro-based system as sustainable alternative for urban logistics in the era of e-commerce. *Sustainability* 13, 4479. <https://doi.org/10.3390/su13084479>.
- Wang, H., 2019. Routing and scheduling for a last-mile transportation system. *Transp. Sci.* 53, 131–147. <https://doi.org/10.1287/trsc.2017.0753>.
- Wang, Y., Zhang, D., Liu, Q., Shen, F., Lee, L.H., 2016. Towards enhancing the last-mile delivery: an effective crowd-tasking model with scalable solutions. *Transp. Res. E Logist. Transp. Rev.* 93, 279–293. <https://doi.org/10.1016/j.tre.2016.06.002>.
- Wei, H., Li, A., Jia, N., 2020. Research on optimization and design of sustainable urban underground logistics network framework. *Sustainability* 12, 9147. <https://doi.org/10.3390/su12219147>.
- Yang, J., Guo, J., Ma, S., 2016. Low-carbon city logistics distribution network design with resource deployment. *J. Clean. Prod.* 119, 223–228. <https://doi.org/10.1016/j.jclepro.2013.11.011>.
- Yu, V.F., Jodiawan, P., Hou, M.-L., Gunawan, A., 2021. Design of a two-echelon freight distribution system in last-mile logistics considering covering locations and occasional drivers. *Transp. Res. E Logist. Transp. Rev.* 154, 102461 <https://doi.org/10.1016/j.tre.2021.102461>.
- Zhan, X., Szeto, W.Y., Wang, Y., 2023. The ride-hailing sharing problem with parcel transportation. *Transp. Res. E Logist. Transp. Rev.* 172, 103073 <https://doi.org/10.1016/j.tre.2023.103073>.
- Zhang, H., Lv, Y., Guo, J., 2022. New development direction of underground logistics from the perspective of public transport: a systematic review based on scientometrics. *Sustainability* 14, 3179. <https://doi.org/10.3390/su14063179>.
- Zhao, L., Li, H., Li, M., Sun, Y., Hu, Q., Mao, S., Li, J., Xue, J., 2018. Location selection of intra-city distribution hubs in the metro-integrated logistics system. *Tunnelling Undergr. Space Technol.* 80, 246–256. <https://doi.org/10.1016/j.tust.2018.06.024>.
- Zhao, L., Zhou, J., Li, H., Yang, P., Zhou, L., 2021. Optimizing the design of an intra-city metro logistics system based on a hub-and-spoke network model. *Tunnelling Undergr. Space Technol.* 116, 104086 <https://doi.org/10.1016/j.tust.2021.104086>.
- Zheng, C., Zhao, X., Shen, J., 2020. Research on location optimization of metro-based underground logistics system with voronoi diagram. *IEEE Access* 8, 34407–34417. <https://doi.org/10.1109/access.2020.2974497>.

- Zheng, C., Zhang, C., Ma, J., Wu, F., Sun, K., 2022. Location selection of metro-based distribution nodes for underground logistics system with bi-level programming model. *Symmetry* 14, 2411. <https://doi.org/10.3390/sym14112411>.
- Zhou, M., Zhao, L., Kong, N., Campy, K.S., Xu, G., Zhu, G., Cao, X., Wang, S., 2020. Understanding consumers' behavior to adopt self-service parcel services for last-mile delivery. *J. Retailing Con. Serv.* 52, 101911.
- Zuo, T., Li, B., Zhang, F., Yin, Y., 2023. Analysis of a collaborative transport model mixing passengers with freights in metro system. *J. Rail Transp. Plan. Manag.* 25, 100358 <https://doi.org/10.1016/j.jrtpm.2022.100358>.